Journal of Early Intervention
Volume 30 Number 4
September 2008 295-320

10.1177/1053815108320814
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The Early Intervention Parenting Self-Efficacy Scale (EIPSES)

Scale Construction and Initial Psychometric Evidence

Amy B. Guimond
M. Jeanne Wilcox
Suzanne G. Lamorey
Arizona State University, Tempe

The psychometric properties of an instrument designed to measure parenting efficacy within the context of early intervention, the Early Intervention Parenting Self-Efficacy Scale (EIPSES), were explored. One hundred seventeen caregivers of children receiving early intervention services completed the 20-item EIPSES. The scale was reduced to 16 items with an internal reliability coefficient of .80. Preliminary factor analyses revealed a 2-dimensional structure for the EIPSES, one related to Parent Outcome Expectations and a second reflecting Parent Competence, together accounting for 37% of the variance. The Parent Outcome Expectations factor was conceptualized as a measure of parents' beliefs in the role of environmental influences, such as early intervention, on children's development. The Parent Competence factor was conceptualized as parents' beliefs in their abilities to promote children's developmental outcomes. Subscale reliability analyses and correlations among related constructs provide initial support for the utility of the EIPSES in assessing task-specific early intervention–related parental self-efficacy.

Keywords: *early childhood intervention; self-efficacy; parents; developmental disabilities*

Individualized family service plans are designed to define successful child and family early intervention outcomes among families of children with disabilities and to delineate the services considered necessary to achieve those outcomes. Highlighting the role of families in the promotion of children's optimal social, emotional, and developmental outcomes has become an emphasis in current research on how best to serve infants and young children with disabilities (Bailey & Bruder, 2005). One such means for promoting positive child and family early intervention outcomes is parental self-efficacy (PSE). Of the many domains of self-efficacy, PSE, or a parent's confidence and competence in parenting, has been gaining recognition as a family outcome of early intervention (Bailey et al., 1998) and

Authors' Note: Lamorey is now affiliated with the University of North Carolina, Charlotte. This study was supported by a grant from the U.S. Department of Education, Award No. R305T000049GR, for which M. Jeanne Wilcox was principal investigator. Official endorsement of statements expressed in this article has not been granted by the U.S. Department of Education nor should it be inferred. Correspondence concerning this article should be addressed to Amy B. Guimond, PhD, Infant Child Research Programs, Arizona State University, P.O. Box 871908, Tempe, AZ 85287-1908; e-mail: amy.guimond@asu.edu.

a modifiable construct through which child outcomes can be achieved (see Jones & Prinz, 2005, for a review).

To influence child abilities through intervention, interventionists are called upon to alter the environment in which the child lives or the capacities of the primary caregiver (Bailey, Hebbeler, Scarborough, Spiker, & Mallik, 2004; Bronfenbrenner, 1979; U.S. Department of Education, 2003). The current investigation is focused on a new scale, the Early Intervention Parenting Self-Efficacy Scale (EIPSES), as a means to measure the process through which caregiver capacity may be altered and, thus, positive child outcomes might be facilitated. Consistent with ecological perspectives, child development is influenced by multiple contexts, including the parent, the early intervention setting, and the larger socio-cultural context (Bronfenbrenner, 1979). Parents' competence-related beliefs are conceptualized here, then, as a context in which child development can be promoted.

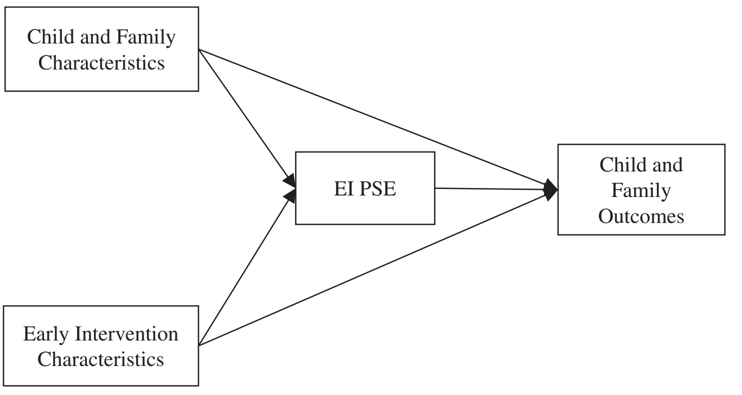
There are available scales that focus on PSE for populations of parents of children without disabilities (e.g., Coleman & Karraker, 2003; Johnston & Mash, 1989; Teti & Gelfand, 1991) as well scales that focus on parenting confidence related to promoting specific child skills (e.g., development of children's language; DesJardin, 2003, as cited in DesJardin, 2005). To our knowledge, however, no scales are designed specifically to assess the early intervention–related competence beliefs of parents of infants and toddlers with a variety of disabilities.

Self-Efficacy

Self-efficacy refers to an individual's perceptions of himself or herself as competent in a given task or domain and is thought to develop through a complex path contingent on an individual's tendency to set and follow goals, face challenges, and recover from failure (Bandura, 1977, 1986, 1997). An important feature of self-efficacy is that it can be domain or task specific and vary from one behavior or context to the next. PSE beliefs, then, are often noted as parental perceptions of the capability to be effective in the domain of parenting and confidence in the ability to perform parenting tasks. Our aim is to define and describe a task-specific form of PSE, namely, early intervention parental self-efficacy (EI PSE). Through this task-specific measure, we examine PSE as it relates to confidence and effectiveness in promoting child development when children are also involved in early intervention services. In addition, we plan to measure EI PSE as it relates to parents' beliefs in the role of environmental or contextual influences on children's development.

Although domain-specific (rather than domain-general) self-efficacy scales, such as PSE, are valuable tools for measuring refined behavior areas, task-specific measures are even more detailed than domain-specific measures of self-efficacy. Furthermore, task-specific measures might provide integral information as to how confidence in specific parenting behaviors (e.g., ability to work with early interventionists to promote child development) are related to changes in parents' abilities to accomplish those specific tasks (Bandura, 1986). It is possible that mothers might feel competent regarding general parenting tasks but not early intervention task-specific skills, such as finding and accessing services for their child. These more refined constructs seem to be an important next step in understanding the relation between beliefs and behaviors.

Figure 1
Conceptual Model of the Theoretical Path Between Early Intervention Parental Self-Efficacy (EI PSE) and Child and Family Outcomes



In Porter and Hsu's (2003) work involving maternal self-efficacy during the transition to parenthood, they suggested that global maternal self-efficacy beliefs tend to be stable. Alternatively, they proposed that discrete, task-specific beliefs are more likely to change over time. It is these task-specific beliefs that might be an optimal avenue to pursue in early intervention settings. Although we discuss PSE primarily as a family outcome to assess early intervention, it is also important to recognize its influence on other aspects of parental adjustment (e.g., stress, depression) and child outcomes (see Figure 1 for a conceptual model). For instance, in a sample of families of children with autism, Hastings and Brown (2002) found that self-efficacy acted as a moderator in the relation between child disorder severity and parental anxiety and depression. In other research among children with and without disabilities, parents' self-efficacy mediated the relation between child characteristics and both parent behavior and well-being (e.g., Donovan, Leavitt, & Walsh, 1990; Halpern & McLean, 1997; Hastings & Brown, 2002; Pit-ten Cate, Kennedy, & Stevenson, 2002). Taken together, it seems that PSE and other child and family characteristics interact in complex ways and evolve in a transactional nature (Bandura, 1997).

Linking PSE With Parenting Behaviors

The study of domain- and task-specific measures of PSE has become an important area of research because of its potential influence on both concurrent and future parenting behaviors. Perceived self-efficacy is, in part, an outcome of experiences, and thus, a parent's prior perceptions of competence in the parenting role are thought to contribute to both current and future perceived PSE as well as actual parenting behaviors (Bandura, 1986). With respect to parenting, efficacy theorists have suggested that increased feelings of effectiveness as a caregiver will lead to enhanced quality of the parent–child relationship (e.g., Bailey et al., 1998).

Among parents of children without disabilities, mothers with high domain- and task-specific PSE tend to be more responsive (Unger & Wandersman, 1985), have greater parental satisfaction (Coleman & Karraker, 2000; Hudson, Elek, & Fleck, 2001), and have children with easy temperaments (Grolnick, Benjet, Kurowski, & Apostoleris, 1997). Conversely, mothers with low task-specific PSE tend to be less sensitive and more awkward during parent–child interactions (Teti & Gelfand, 1991). In studies using domain-general measures of PSE, mothers tend to perceive more child behavior problems and difficulty (Cutrona & Troutman, 1986; Johnston & Mash, 1989) and are less able to manage difficult child behaviors (Gross, Fogg, & Tucker, 1995). Although the majority of studies involving PSE have examined it as a correlate or predictor of parenting behaviors, it is likely that the relation between parenting beliefs and behaviors is reciprocal in nature.

Across studies of both families of children with and without disabilities, low domain- and task-specific self-efficacy has been related to stress and depression (Cutrona & Troutman, 1986; Hastings & Brown, 2002; Machida, Taylor, & Kim, 2002) as well as family dysfunction (Scheel & Rieckmann, 1998) and marital conflict (Teti & Gelfand, 1991). In studies of children with disabilities, these relations seem particularly pertinent because parental competence and adjustment may already be at risk because of increased incidence of depression among parents (Breslau & Davis, 1986) and the added stressors often associated with raising a child with challenging caretaking issues (Fisman, Wolf, & Noh, 1989; Sanders & Morgan, 1997).

Interventions aimed at promoting parents' cognitive appraisals of their own parenting abilities and their children's attributes and behaviors may have positive effects on outcomes for the parent, child, and family as a whole. Among families of children with disabilities, family-centered programs have been linked to positive child outcomes when parents reported positive appraisals of personal control and support (Dunst, 1999). Furthermore, intervention practices that target parental empowerment have been shown to positively influence parents' self-efficacy beliefs regarding perceived control over life circumstances as well as general parenting practices (e.g., Judge, 1997; Miller-Heyl, MacPhee, & Fritz, 1998; Spoth, Redmond, Haggerty, & Ward, 1995). Targeting PSE through parent behavioral training also has been shown to be effective in increasing positive parenting behaviors, parenting self-efficacy, and decreasing parental stress immediately following training and continuing through a follow-up 1 year later (Gross et al., 1995; Tucker, Gross, Fogg, Delaney, & Lapporte, 1998).

Jones and Prinz's (2005) review of literature highlights both the complexity of the relations between PSE and child and parent adjustment as well as the importance of acknowledging context in these relations. Specifically, they noted that variability among parents, children, and culture might be the key to determining the strength and direction of the links between PSE and parenting and child outcomes. Their focus on context indicates the importance of studying PSE as a family outcome with parents of children with disabilities in addition to other demographic factors.

Investigators have examined the relations between PSE scales and a variety of demographic and family characteristics (i.e., parent and child age, number of children in the household, income, and education). Across studies of children without disabilities (see Coleman & Karraker, 1998, or Jones & Prinz, 2005, for a review), some consistent patterns have emerged between PSE and family characteristics. Few researchers, however, have examined these relations with families of children with disabilities.

Researchers have suggested that sociocontextual characteristics, as they create heightened levels of stress in the family, most likely indirectly affect PSE (cf. Coleman & Karraker, 1998). For instance, Coleman and Karraker (2000) measured both domain-general and domain-specific PSE and found that both measures were positively and moderately related to maternal education and family income. Relations between PSE and sociocontextual characteristics may also depend on the age of children as well as the specificity of the self-efficacy scale used (i.e., domain general, or domain specific or task specific). Therefore, our present research explores the relations between EI PSE and a variety of family and demographic factors, and it may show whether those relations exist with a task-specific PSE scale in a sample of parents of young children with developmental delays.

Linking PSE to Child Social, Emotional, and Developmental Functioning

Because researchers have demonstrated that difficult infants are susceptible to parenting difficulties (e.g., Park, Belsky, Putnam, & Crnic, 1997), it is important to understand how parents' perceptions of child behaviors and their competence are linked, particularly within a potentially stressful context of raising a child with a disability. Researchers have noted that socializing a difficult child may lead to lower feelings of PSE across studies of both children with and without disabilities (e.g., Gross, Conrad, Fogg, & Wothke, 1994; Gross & Rocissano, 1988; Leerkes & Crockenberg, 2000; Teti & Gelfand, 1991).

With respect to children's developmental functioning, less is known about its relation to parenting self-efficacy. In samples of children without disabilities, Coleman and Karraker (2003) found statistically significant and positive relations between their domain-specific maternal self-efficacy scale and the mental scale of the Bayley Scales of Infant Development (Bayley, 1993). Similarly, Swick and Hassell (1990) determined that parents' self-efficacy was positively related to children's overall developmental functioning. Within populations of children with disabilities, the relation between parenting self-efficacy and child social, emotional, and in particular, developmental outcomes deserves more attention.

Extant data indicate that the relation between child characteristics and PSE is a function of difficult child behaviors and temperament (e.g., Hastings & Brown, 2002; Mash & Johnston, 1983; Teti & Gelfand, 1991). For instance, researchers have found that parents of children who are hyperactive tend to have lower levels of self-efficacy when the child also had a difficult temperament (Mash & Johnston, 1983; Teti & Gelfand, 1991). Reasoning that because parents may not understand, or may become frustrated with, their children's difficult behavior, the authors suggested that parents might feel that they have less influence or effectiveness on their child's development. Furthermore, in their sample of parents of children with autism, Hastings and Brown (2002) found that self-efficacy mediated the influence of child behavior problems on anxiety and depression for mothers and moderated the effect of child behavior problems on anxiety for fathers. Because of these findings, an additional component of this investigation was to explore the relations between children's social, emotional, and developmental functioning and the EIPSES.

EIPSES

The development of the EIPSES was based on existing concerns that few researchers have explored the role of PSE in families of children with disabilities (Gowen, Johnson-Martin, Goldman, & Appelbaum, 1989; Hastings & Brown, 2002; Pit-ten Cate et al., 2002; Scheel & Rieckmann, 1998). As such, there is a limited number of tools available with which to study task- or domain-specific self-efficacy among a population of parents raising children with disabilities. PSE scales that do exist for this population are designed for particular disability groups, such as children with Asperger syndrome (Sofronoff & Farbotko, 2002), prelingual deafness (DesJardin, 2005), or seizure disorders (Caplin, Austin, Dunn, Shen, & Perkins, 2002). As a result, researchers focused on parents of children with disabilities can rely only on general PSE measures that were designed for parents of children without disabilities.

To our knowledge, few existing instruments allow professionals to focus on understanding whether parents feel competent and confident in their skills, knowledge, and the ability to make a difference in the lives of their children, especially when their infant or toddler also participates in early intervention services because of risk or developmental delay. A scale with this emphasis is important because involvement with early intervention programs may provide challenges to parents who are raising children with or at risk for developmental delays. Moreover, existing general parenting efficacy scales may not capture needed information.

The measure developed in the present study focused on early intervention parenting self-efficacy. It was developed to address the need for viable measures of change in families who are receiving early intervention. The study was guided by Guralnick's (1998) proposition that a primary factor in determining successful from unsuccessful interventions lies in the determination of the degree to which parents feel competent and confident in parenting their child with or at risk for disability. In the present investigation, we explore the utility and initial psychometric properties of the EIPSES as a measure of early intervention PSE beliefs. The measure was developed with the intent to quantify parent perspectives about their ability to facilitate positive child outcomes within the context of early intervention programs and via interactions with early intervention practitioners.

Theoretical Basis of the EIPSES

The EIPSES was largely based on an existing scale designed to assess teacher self-efficacy (Gibson & Dembo, 1984). A teaching efficacy scale was used as a basis for the EIPSES with the expectation that parents of children with disabilities often take on a "teaching" role as they manage and participate in their child's early intervention services. Gibson and Dembo (1984) derived two dimensions of teacher efficacy from their scale, general teaching efficacy (IE) and personal teaching efficacy (PE), both of which are similar to Bandura's (1997) notions of outcome expectations and efficacy expectations, respectively. For example, mapping Bandura's ideas of outcome expectancies onto the notion of TE beliefs is indicative of the degree to which a parent believes that child outcomes are a function of environmental influences or constraints, such as family background or the availability of early intervention or community support. Conversely, in relating efficacy expectations to the PE dimension, this translates into parental beliefs of being personally capable of producing positive changes in their child and of their ability to promote optimal child development. With respect to the EIPSES, we expected that items might map similarly onto two dimensions of parenting efficacy—one related to general outcome expectations (Parent Outcome Expectations) and one related to personal efficacy expectations (Parent Competence).

Method

Participants

The data reported in this article were collected as part of a larger study (Wilcox & Lamorey, 2004) involving relationship-based training practices in early intervention. During the course of 1 year, information was collected from primary caregivers and their infants and toddlers who were receiving early intervention services from local area agencies in Arizona, Utah, and northern California. Participants selected for the present study were all receiving services from early interventionists through Part C of the Individuals With Disabilities Education Act (IDEA) Amendments of 1997.

One hundred seventeen infants and toddlers (72 boys, 45 girls) and their primary female caregivers (112 biological mothers, 5 adoptive mothers) completed a baseline EIPSES and were included in the final sample. Fathers ($n = 7$) and grandmothers ($n = 1$) were excluded from the study because of insufficient sample size to make between-group comparisons and because researchers have suggested that the relation between child and parenting self-efficacy variables may be different for mothers and fathers (e.g., Hastings & Brown, 2002; Johnston & Mash, 1989).

Children ranged in age from 3 months to 34 months ($M = 17.28$, $SD = 6.82$), and caregivers' ages ranged from 16 years to 52 years ($M = 31.11$, $SD = 6.99$), at the time of baseline data collection. Child disability conditions and diagnoses were obtained via caregiver report. Caregiver reports of child disability diagnoses included Down syndrome ($n = 14$), physical disability ($n = 23$), epilepsy ($n = 8$), vision impairment ($n = 18$), cerebral palsy ($n = 11$), speech and/or hearing impairment ($n = 11$), chronic illness ($n = 4$), mental retardation ($n = 9$), nonspecified developmental delay ($n = 75$), and other delay or disability ($n = 24$). Parents were able to select more than one disability category, so many children were reported as having multiple or comorbid disabilities or delays. Although specific diagnoses were not verified by additional sources, all children were qualified for Part C services in their respective states.

Table 1 shows child, family, and demographic characteristics of the sample. According to caregiver report of ethnicity, 23.1% of caregivers were Hispanic or Latino, 65.8% were White, and the remaining 11.1% were of other ethnic background. A minority of the primary caregivers had received less than a high school education (5.2%) or had a high school diploma or the equivalent (23.5%), whereas 27% of the sample had received at least some college education, 9.6% had received a community college degree, and one third of the caregivers had received a college degree or higher. Most of the children (88.9%) were from two-parent households. Approximately 17.7% of the sample had an annual family income of $15,000 or less, 28% had an annual income between $15,001 and $35,000, 42.9% had

Table 1
Means and Standard Deviations of the Early Intervention Parenting Self-Efficacy Scale (EIPSES) Total and Factor Scores for Total Sample and by Demographic Subgroups

		Total EIPSES		Parent Outcome Expectations		Parent Competence	
Group	*n*	*M*	*SD*	*M*	*SD*	*M*	*SD*
Child gender							
Boys	72	5.75	.64	5.68	.79	5.90	.79
Girls	45	5.94	.65	5.91	.73	6.02	.73
Ethnicity							
White	77	5.92	.59	5.93	.64	5.89	.74
Hispanic	27	5.65	.79	5.44	1.03	6.07	.74
Other	13	5.65	.55	5.51	.64	6.02	.69
Mother’s education							
No high school diploma	6	5.67	.72	5.45	.82	6.21	1.14
High school diploma only	27	5.80	.66	5.71	.82	6.05	.66
Some post–high school	31	5.96	.65	5.97	.75	5.93	.75
Associate’s degree	11	5.25	.45	5.08	.52	5.68	.60
Bachelor’s degree	24	5.84	.66	5.90	.73	5.70	.81
Graduate degree	15	6.04	.56	5.97	.77	6.23	.48
Child disorder severity							
Mild	46	5.97	.63	5.92	.75	6.11	.70
Moderate	44	5.76	.64	5.71	.75	5.89	.79
Severe	11	6.05	.54	6.05	.56	6.05	.76
Yearly household income							
<$8,000	10	5.60	.69	5.36	.89	6.18	.64
$8,000 to $15,000	9	5.75	.55	5.67	.70	5.94	.58
$15,000 to $25,000	13	5.66	.81	5.67	.94	5.65	.86
$25,000 to $35,000	17	5.95	.79	5.91	.84	6.06	.97
$35,000 to $45,000	12	5.92	.57	5.96	.71	5.83	.79
$45,000 to $55,000	12	5.86	.55	5.89	.55	5.77	.79
$55,000 to $65,000	10	5.70	.54	5.61	.60	5.93	.59
$65,000 to $75,000	8	6.14	.67	6.11	.74	6.22	.71
>$75,000	16	6.02	.40	6.09	.50	5.84	.56

between \$35,001 and \$75,000, and 7.5% had more than \$75,001. All children were living with their biological or adoptive caregivers.

Procedures

The EIPSES was administered three times as part of an ongoing investigation (Wilcox & Lamorey, 2004) aimed at identifying the associations between relationship-based practices in early intervention and child and family outcomes. As part of the larger study, 101 early intervention practitioners were recruited from local agencies in the states of Arizona, Utah, and northern California. Family participants for the larger data set (N = 162) included those with children with or at risk for developmental disabilities, younger than 3 years of age at the time the investigation began, and who were currently receiving intervention services on a regular basis through Early Head Start or federal Part C programs.

Upon initial recruitment, practitioners were randomly assigned to treatment and contrast groups. Practitioners in the treatment group received training related to the implementation of relationship-based practices in early intervention, and those in the contrast group received no training. For the current study, baseline data of families receiving Part C services were used to assess the psychometric properties of the EIPSES before any intervention training of the practitioners took place.

Practitioners were asked to identify two families who were currently receiving their services, would be younger than 36 months before the end of the study, and were interested in participating. If families refused participation, practitioners were asked to select an alternative family. Although study investigators requested that each early interventionist recruit two families to take part in the study, some practitioners (n = 35) were able to solicit participation from only one family they were currently serving. No other specific instructions were given to practitioners regarding family recruitment. Once practitioners identified families interested in participating in the study, a graduate research assistant (RA) contacted those families to set up an appointment in the family's home to review study procedures and obtain informed consent. During the same home visit, one RA administered a questionnaire packet to the caregiver while another RA conducted a developmental assessment of the child. Questionnaire packets included a family demographic form, a questionnaire related to the child's social and emotional development, and the EIPSES. Each family was paid \$15 at the time of the home visit. All measures were available in both Spanish and English, and bilingual RAs conducted home visits for Spanish-speaking families.

Measures

Demographics. Within the demographic portion of the questionnaire, caregivers answered questions regarding their level of education, family income, and the number of children and adults living in the home. Additionally, caregivers self-reported on both their and their child's race or ethnic origin, gender, and age. Parents also self-reported on their child's disability category and severity of disability (i.e., mild, moderate, or severe).

The EIPSES. The EIPSES purports to assess (a) the degree to which caregivers perceive of themselves as being personally effective and capable in parenting their child and (b) the extent to which they believe child outcomes are a function of environmental influences or constraints, such as family background or the availability of early intervention or community support. The EIPSES was adapted, in part, from a measure of teacher self-efficacy, the Teacher Efficacy Scale (TES; Gibson & Dembo, 1984). The TES is a 30-item scale using a 6-point Likert-type format, with higher scores indicating higher levels of teacher efficacy. Factor analysis of the original 30-item TES scale resulted in two subscales, Personal and General Teaching Efficacy, with internal consistency reliability coefficients of .78 and .75, respectively.

Based on recommended practices in early intervention, 12 items of the TES were revised to reflect the specific situational context of parents' roles and responsibilities as members of their child's early intervention team (see Table 2 for items). Six items each were selected to represent the personal and general self-efficacy domains identified in the TES, respectively. Statements for teachers on the TES were modified for parents so that the items corresponded to parents' beliefs in their ability to participate meaningfully in affecting child outcomes under the provision of early intervention services. In addition, eight new items were written to address areas that the TES did not cover. These items reflected the influence of perceived collaboration with early interventionists, environmental influences and community resources, and parenting experiences.

The resultant EIPSES consisted of 20 items using a 7-point Likert-type scale for responses ranging from *strongly disagree* (1) to *strongly agree* (7). Total scores were computed by summing all items of the scale. Scoring for items 3, 5, 6, 8, 12, 16, 17, 19, and 20 were reversed so that for all items, higher scores reflected greater perceived self-efficacy.

Child measures. Several baseline measures focusing on child behaviors and developmental functioning were examined to assess their relations with the EIPSES. One of the child measures was the Battelle Developmental Inventory (BDI; Newborg, Stock, Wnek, Guidubaldi, & Svinicki, 1988), a standardized norm-referenced assessment designed to measure personal–social, adaptive, motor, communication, and cognitive development of children from birth to 8 years. In the current research study, the BDI was administered by qualified graduate research assistants. The overall BDI developmental quotient (DQ) score ($M = 100$, $SD = 15$), as well as each of the domain DQ scores, was retained.

The second baseline child measure used was the Infant-Toddler Social and Emotional Assessment (ITSEA; Carter & Briggs-Gowan, 2000). The ITSEA is a parent-report instrument assessing the social and emotional development of infants and toddlers and measures the following four domains of behavior: internalizing, externalizing, dysregulation, and competence. The internalizing domain includes the scales Depression–Withdrawal, General Anxiety, Separation Distress, and Inhibition to Novelty. Sample items for the internalizing domain include "seems nervous, tense or fearful," "does not make eye contact," and "gets upset when left with a familiar babysitter or relative." The externalizing domain includes the Activity–Impulsivity, Aggression–Defiance, and Peer Aggression scales and includes items such as "is restless and can't sit still" and "hits, shoves, kicks, or bites other children." The dysregulation domain comprises Sleep, Negative Emotionality, Eating, and Sensory Sensitivity scales. Sample items in the Dysregulation scale include "sleeps through the night" (reverse scored) and "is whiny or fussy when s/he is *not* tired." The competence

Table 2
Early Intervention Parenting Self-Efficacy Scale (EIPSES) Items Based on Modified Teaching Efficacy Scale (TES) Items

Item Number	Wording
1	If my child is having problems, I would be able to think of some ways to help my child. *TES item: When a student is having difficulty with an assignment, I am usually able to adjust it to his/her grade level.*
2	When my child shows improvement, it is because I am able to make a difference in my child's development. *TES item: When the grades of my students improve, it is usually because I found more effective teaching approaches.*
3	When it comes right down to it, parents really can't do much because most of a child's development depends on their early interventionists. *TES item: A teacher is very limited in what he/she can achieve because a student's home environment is a large influence on his/her achievements.*
4	If one of my child's early interventionists has difficulty with my child, I would be able to offer some suggestions. *TES item: If one of my students could not do an assignment, I would be able to accurately assess whether the assignment was at the correct level of difficulty.*
5	Children will make the most progress if their early interventionists work with them rather than if the parents work with the children. *New item: Designed to examine parents' beliefs regarding the role of early interventionists in promoting child progress.*
6	Even a good parent may not have much impact on whether children feel good about themselves. *TES item: Even a teacher with good teaching abilities may not reach many students.*
7	I feel that I can work well with my child's early interventionist as part of my child's team. *New item: Designed to reflect parents' beliefs in the ability to collaborate in a family-centered approach to early intervention.*
8	Because there is so little help from the community, I am often sad or angry about how few services I can find for my child and the rest of my family. *TES item: Many teachers are stymied in their attempts to help students by a lack of support from the community.*
9	If my child learns something quickly, it would probably be because I know how to help my child learn new things. *TES item: If a student masters a new math concept quickly, this might be because I knew the necessary steps in teaching that concept.*
10	If my early interventionist did not have some information we had talked about before, I would be able to find the information and pass it on to my early interventionist. *TES item: If a student did not remember information I gave in a previous lesson, I would know how to increase his/her retention in the next lesson.*
11	When my child does better than expected, many times it is because I tried a little harder to help. *TES item: When a student does better than usual, many times it is because I exerted a little extra effort.*
12	The amount that a young child will learn is mostly due to family background, the neighborhood, and the early interventionist rather than their parents. *TES item: The amount that a student can learn is primarily related to family background.*

(continued)

Table 2 (continued)

Item Number	Wording
13	I could help my child much more if the rest of the world were more understanding. *New item: Designed to explore the influence of environmental/community resources on parents' EI self-efficacy.*
14	Overall, the help that I give to my child's early interventionists has a big effect on my child. *New item: Designed to explore the influence of collaboration on parents' EI self-efficacy.*
15	On most days, I can handle most of the ups and downs of being a parent. *New item: Designed to reflect the role of environmental influences on parents' EI self-efficacy.*
16	I worry that I am not a good enough parent due to outside demands placed upon my time and energy. *New item: Designed to explore the role of environmental influences on parents' EI self-efficacy.*
17	When my child is ill, I feel that there is nothing I can do to help my child or other members of my family. *New item: Designed to reflect the role of environmental influences on parents' EI self-efficacy.*
18	Over the past year, I can see the progress that I have made in becoming a better parent. *New item: Designed to reflect the role of experience on parents' EI self-efficacy.*
19	No matter how hard I try, it seems that I just cannot find a way to get the services that my child and my family needs. *New item: Designed to explore the role of community resources on parents' EI self-efficacy.*
20	The traits that a child has before he or she is born are more important than anything that the child's parents can do for the child. *TES item: Teachers are not a very powerful influence on student achievement when all factors are considered. Modification was made to reflect the role of genetic predisposition on parents' EI self-efficacy.*

Note: EI = early intervention.

domain includes the scales Compliance, Attention, Imitation–Play, Mastery Motivation, Empathy, and Prosocial Peer Relations and includes items such as "tries to make you feel better when you are upset" and "pretends that objects are something else."

The ITSEA has been used with infants and toddlers without disabilities (e.g., Carter & Briggs-Gowan, 2000), children with autism (e.g., Mahoney & Perales, 2003), and children with chronic health and developmental problems (Horwitz, Gary, Briggs-Gowan, & Carter, 2003). Internal consistency alphas for each of the four scales were .92 for Externalizing, .75 for Internalizing, .78 for Dysregulation, and .84 for Competence in a pediatric sample of 214 parents of infants and toddlers (Briggs-Gowan & Carter, 1998). Because the ITSEA was designed specifically for use with infants and toddlers ages 11 to 36 months, and only has standard scores available for this age range, only a subset of our current sample was used in analyses involving the ITSEA.

Results

In this section, we present preliminary psychometric properties of the EIPSES, results from exploratory factor analyses, subsequent scale modifications, and score reliability estimates for a proposed factor structure of the EIPSES. Next, we discuss descriptive information on the EIPSES total and factor scores with respect to a set of demographic characteristics. Finally, we provide initial evidence in support of the construct validity of EIPSES scores by correlating scores on the EIPSES with measures reflecting children's developmental, social, and emotional functioning.

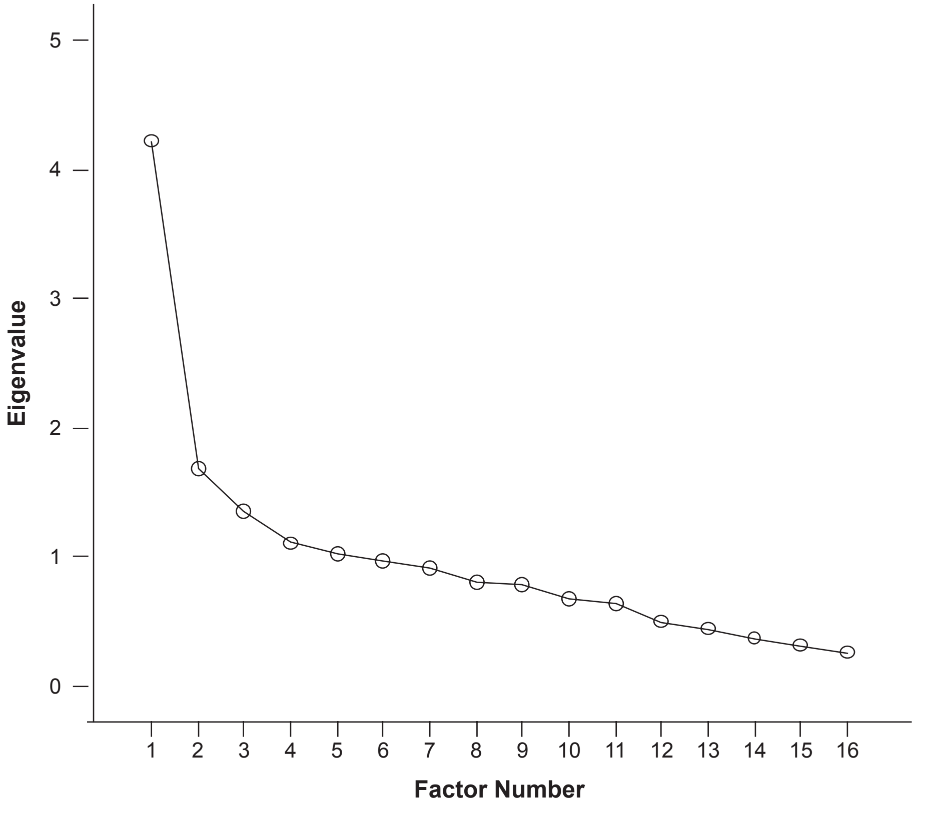
Psychometric Properties of the EIPSES

First, internal consistency score reliability for the 20-item scale was evaluated with a resulting Cronbach's alpha coefficient of .81. Means for the 20-item scale ranged from 4.95 to 6.24, indicating small to moderate variance in responses for the 1- to 7-point Likert-type scale. Because of the criterion that items with item-to-total correlations of .30 or less be deleted from the scale to ensure adequate score reliability, 4 items were deleted, resulting in a 16-item scale (cf. Nunnally, 1978; Nunnally & Bernstein, 1994). As suggested by Ferketich (1990), item–total correlations were also examined; however, no item–scale correlations exceeded a .70 threshold indicating redundancy of those items. Internal consistency score reliability of the subsequent 16-item scale was .80, and means ranged from 4.94 to 6.70, again indicating a small to moderate amount of variance in item responses.

An exploratory factor analysis was conducted on the remaining 16 items of the EIPSES to extract underlying factors in the structure of the scale (Munro, 2001). Although statisticians generally agree that when conducting factor analyses, larger sample sizes are better for obtaining parameter estimates that reflect the population, they vary in their recommendations of sample size guidelines. Several rules of thumb indicate that our sample was acceptable in that there were at least five participants per measured variable (Gorsuch, 1983) and that the sample size was greater than 100 (Hatcher, 1994; MacCallum, Widaman, Zhang, & Hong, 1999). In addition, the Kaiser-Meyer-Olkin statistic of sample adequacy was .72, which exceeds generally accepted levels of .60, and further supported the use of a factor analysis (Kaiser & Rice, 1974).

Fabrigar, Wegener, MacCallum, and Strahan (1999) suggested that principal axis factoring is recommended rather than principal components methods when one's data violate the assumption of normality. Because of the restricted variance in the data and non-normal distribution, a principal axis factoring method was used, with prior communalities estimated using squared multiple correlations (Gorsuch, 1983). Multiple criteria were considered when determining the number of factors to retain (a) the a priori hypothesis that the measure had two dimensions, (b) Kaiser's (1970) "eigenvalues greater than 1" criterion, (c) the scree test (Cattell, 1966), and (d) the interpretability of the factor solution. After initial factoring of the scale (unrotated), alpha extraction resulted in five factors with eigenvalues greater than 1 (Kaiser, 1970), accounting for 58.6% of the total variance. Because some researchers have reported that the Kaiser criterion can result in an overestimate of factors (e.g., Lance, Butts, & Michels, 2006), Cattell's (1966) scree plot test was administered next to establish a more parsimonious analysis. Consistent with Cattell's test, which suggests excluding any factors that fall below the "elbow" of the scree plot, two factors emerged (see Figure 2). Taking into account the Kaiser and Cattell criteria, four-, three-, and two-factor solutions were examined further.

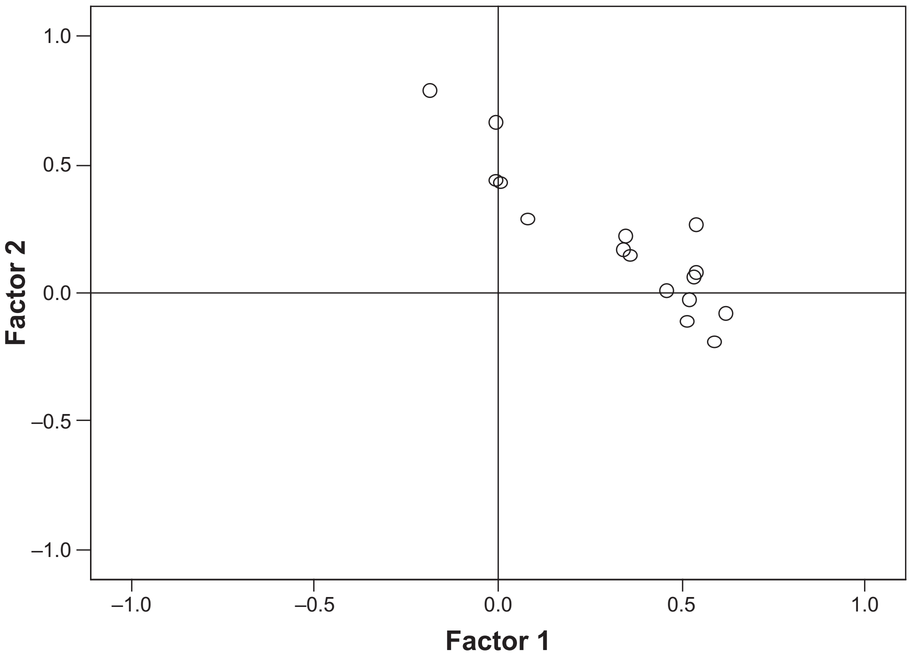
Figure 2
Scree Plot of Early Intervention Parental Self-Efficacy Scale (EIPSES) Eigenvalues by Factor



Although an orthogonal varimax rotation tends to provide clearer results, it is more accurate to use oblique methods when correlated factors are assumed. Furthermore, Preacher and MacCallum (2003) suggested oblique rotations be used in lieu of orthogonal rotations, as orthogonal solutions more accurately depict the typical nonindependent nature of multidimensional constructs. Because of this information, both a varimax and promax oblique rotation were used to aid in the interpretability of results.

Osborne and Costello (2005) argued that the rotation with the "cleanest" factor structure should be retained. The two-factor solution provided the most interpretable results as compared to the four- and three-factor solutions, which did not clearly delineate comprehensible factor structures. The two-factor structure using a promax oblique rotation, in this case, appeared to be the best choice. The amount of total common variance accounted for decreased by 5% going from the three- to the two-factor solution and an additional 4% from the from-factor solution. Compared to the varimax rotation, the promax oblique rotation had fewer items with no or cross-loading factor coefficients and had at least three items in

Figure 3
Rotated Factor Plot for the Early Intervention Parental Self-Efficacy Scale (EIPSES) Factors 1 and 2 (Promax Oblique Rotation)



each factor. The promax oblique rotation with kappa set to the default value of 4 (SPSS, 2006) revealed a correlation between the factors of $r = .50$. Kappa values indicate the extent to which factors are allowed to correlate in the factor analysis (Osborne & Costello, 2005). Figure 3 shows a schematic of the rotated factors using a promax oblique rotation.

Alpha extraction of the two factors followed by promax oblique rotation resulted in factor structure coefficients ranging from .10 to .70 (see Table 3 for pattern and structure coefficients). The first extracted factor accounted for 26% of the variance in the scale, whereas the second factor accounted for an additional 11% of the variance. One item did not meet the established criterion of having a postrotated factor structure coefficient at or greater than .40 (i.e., “On most days, I can handle the ups and downs of being a parent.”). Furthermore, one item had factor structure coefficients greater than .40 on both factors (i.e., “When my child is ill, I feel that there is nothing I can do to help my child or other members of my family.”).

After examination of the two factors, the first factor (Factor 1) comprised 10 items related to parents’ perceptions of their ability to control early intervention outcomes for their children. Of these items, two did not reach coefficient levels greater than .40 for the pattern matrix but did reach these levels for the structure matrix (see Table 2). Because values in the pattern matrix can sometimes be suppressed because of correlations between factors (Field, 2005), these items were retained in Factor 1. These items focused on statements

Table 3
Principal Axis Factoring Factor Coefficients Using a Promax Oblique Rotation (Pattern and Structure Matrices; Kappa = 4)

	Pattern Matrix		Structure Matrix	
Item	Factor 1	Factor 2	Factor 1	Factor 2
12	*.62*	–.08	*.58*	.23
6	*.59*	–.19	*.50*	.10
3	*.54*	.08	*.58*	.35
19	*.53*	.06	*.56*	.32
5	*.52*	–.03	*.50*	.23
20	*.51*	–.11	*.46*	.14
16	*.46*	.01	*.46*	.24
7	*.36*	.15	*.43*	.33
4	*.35*	.22	*.46*	.39
8	*.34*	.17	*.42*	.34
9	–.19	*.79*	.21	*.70*
2	.00	*.66*	.33	*.66*
1	.00	*.44*	.22	*.44*
18	.01	*.43*	.22	*.43*
Deleted item				
15	.08	.29	.22	.33
17	.54	.27	.68	.54

Note: Figures in italics indicate items retained in each factor.

such as "When it comes right down to it, parents really can't do much because most of a child's development depends on their early interventionist to help." Postrotation structure coefficients for items on Factor 1 ranged from .42 to .58. Factor 1 was identified as the Parent Outcome Expectations factor and was similar to Bandura's (1997) theoretical definition of outcome expectancies as well as the TE factor of the TES (Gibson & Dembo, 1984).

The second factor (Factor 2) had postrotation structure coefficients ranging from .43 to .70 and comprised four items associated with parents' perceptions of their parenting abilities and competence. Factor 2 included items related to parents' beliefs about their ability to enhance or affect positive change in the development of their child. For example, "If my child is having problems, I would be able to think of some ways to help my child." Factor 2 is identified as the Parent Competence factor and is similar to Bandura's definition of efficacy expectations and Gibson and Dembo's (1984) PE factor of the TES.

Descriptive Statistics

Total mean scores for each factor were computed by averaging all of the items, with higher scores in each factor reflecting a greater degree of parent competence and outcome expectations. For the current sample, Cronbach's alphas for the Parent Outcome Expectations and Competence factors were .64 and .75, respectively.

Table 4
Associations Between the Early Intervention Parenting Self-Efficacy Scale (EIPSES) Total and Factor Scores and Demographic and Child Variables

	EIPSES Total	Parent Outcome Expectations	Parent Competence
Demographic variables			
1. Child age	–.09	–.06	–.12
2. Mother's age	.04	.09	–.10
3. Number of children	.07	.06	.03
4. Income	.18	.23*	–.03
Battelle Developmental Inventory			
5. Personal–social	–.02	–.03	.02
6. Adaptive behaviors	.07	.06	.07
7. Fine motor	.02	.05	–.06
8. Gross motor	.01	.03	–.04
9. Expressive communication	.13	.14	.04
10. Receptive communication	.24*	.25**	.07
11. Cognitive	.08	.09	.00
12. Total score	.07	.08	.01
ITSEA			
13. Internalizing	–.30**	–.30**	–.15
14. Externalizing	–.29**	–.26*	–.23*
15. Dysregulation	–.31**	–.29**	–.21*
16. Social competence	.16	.10	.22*

Note: ITSEA = Infant-Toddler Social and Emotional Assessment.
* $p < .05$. ** $p < .01$.

The mean scale score for the EIPSES Parent Outcome Expectations and Competence factors were 5.77 (SD = .77) and 5.95 (SD = .74), respectively, for the sample and 5.82 (SD = .64) for the 16-item scale, indicating that most caregivers reported moderate to high levels of self-efficacy. EIPSES mean scale and subscale scores were correlated with and compared across several demographic groups. Zero-order correlations among mean EIPSES scale and subscale scores and child age, caregiver age, and number of children in the household are presented in Table 4. Child age, mother's age, and number of children in the household were not significantly correlated to either of the EIPSES factors or the scale overall.

Means and standard deviations for the EIPSES scale and subscale scores across demographic subgroups (i.e., child gender, mother's ethnicity, mother's education level, mothers' reports of the severity of the child's disability, and family income) are presented in Table 1. Results of one-way ANOVAs revealed no statistically significant differences between boys and girls on the Parent Outcome Expectations or Competence subscale or total scale score. Additionally, no noteworthy associations arose between the EIPSES full scale and subscales with parent ratings of child disorder severity or caregiver reports of yearly household income. Group differences were found across ethnicity subgroups for the Parent Outcome Expectations subscale. Tests of mean differences using a Bonferonni procedure revealed a statistically significant mean difference between Hispanic and White caregivers such that White caregivers rated themselves as having higher mean levels of outcome expectations efficacy than Hispanic caregivers ($p < .01$, $d = .57$). This effect was medium in size according to Cohen's (1988, 1992) effect size conventions. Mean differences also were found across education levels for both EIPSES full scale and Outcome Expectations subscale scores. According to Cohen's conventions, these differences had large effect sizes. Specifically, caregivers with an associate's degree had lower EIPSES total scale scores than caregivers with some post–high school education with no degree or a graduate school degree (*p*s = .03 and .03, *d*s = –1.27 and –1.56, respectively). Similarly, for the Outcome Expectations subscale, caregivers with an associate's degree had lower mean scores than mothers with some post–high school education with no degree, a college degree, or a graduate school degree (*p*s = .01, .04, and .04 and *d*s = –1.34, –1.29, and –1.35, respectively).

Relation of EIPSES to Child Measures

Table 4 delineates correlations among the Parent Outcome Expectations and Competence mean subscale scores. In addition, correlations are shown between EIPSES mean scale scores and child behavior and developmental variables, which have been shown to be correlates of self-efficacy in previous studies. Intercorrelations among these variables revealed that the EIPSES full-scale mean scores were significantly and negatively related to caregivers' ratings on the ITSEA for internalizing ($r^2 = .09$), externalizing ($r^2 = .08$), and dysregulation behaviors ($r^2 = .10$), where lower scores on the EIPSES were related to higher mother-reported internalizing, externalizing, and dysregulation scores. According to Cohen's (1988, 1992) conventions regarding effect sizes of correlations, correlations between the EIPSES scores and ITSEA were small to medium. The EIPSES total score was not related to children's social competence scores on the ITSEA.

EIPSES Parent Outcome Expectations subscale mean scores were significantly and negatively related to caregivers' reports of children's externalizing ($r^2 = .09$) and dysregulation ($r^2 = .07$) behaviors as well as positively related to caregivers' reports of children's social competence behaviors ($r^2 = .08$). Specifically, higher ratings by caregivers of children's externalizing and dysregulation behaviors were related to lower beliefs in perceived outcome expectations. Conversely, higher scores of mothers' perceptions of their child's social competence were related to higher ratings on the Parent Outcome Expectations factor of the EIPSES. Alternatively, the Parent Competence factor of the EIPSES was significantly and negatively related to mothers' reports of children's externalizing ($r^2 = .05$), dysregulation ($r^2 = .04$), and social competence ($r^2 = .05$) behaviors but not children's internalizing behavior scores ($r^2 = .02$). Effect sizes for the relation between the ITSEA subscales and the Parent Outcome Expectations and Parent Competence subscales were small to medium, and small, respectively, according to Cohen's conventions (Cohen, 1988, 1992).

Both the EIPSES total and Parent Outcome Expectations subscale were significantly related to the receptive communication domain of the BDI. The coefficient of determination indicated 6% shared variance between both the EIPSES total and Parent Outcome Expectations subscale with receptive communication. The EIPSES total and subscale scores were not related to any other of the BDI domain scores.

Discussion

Our findings support the EIPSES as a viable task-specific alternative to general parenting self-efficacy measures, suitable for parents of infants and toddlers receiving early intervention services. Results of the factor analyses indicated that the EIPSES measures two aspects of EI PSE: parent outcome expectations and parent competence. It should be noted that the relations between the EIPSES and social emotional and developmental outcomes revealed only low to moderate coefficients of determination, and therefore, the results should be viewed tentatively. Nevertheless, some findings emerge that support previous research relating PSE to child social emotional and developmental characteristics.

In general, both the Parent Outcome Expectations and Competence factors were not related to most BDI (Newborg et al., 1988) scores; nevertheless, they were associated with child measures of social and emotional functioning on the ITSEA (Carter & Briggs-Gowan, 2000). As predicted, both factors of the EIPSES were negatively related to caregiver reports of children's externalizing and dysregulation behaviors. Parent Competence scores, however, but not Outcome Expectation scores, were positively related to parents' ratings of child social competence. Specifically, mothers who perceived they had high levels of parenting competence also tended to report higher levels of social competence in their children. The discrepancies in findings may have been a result of the different characteristics being measured by each of the ITSEA domains. Whereas internalizing, externalizing, and dysregulation domains seem to be salient measures of behavioral or temperamental characteristics of the child (e.g., negative emotionality, activity–impulsivity), social competence relates to measures of specific developmental tasks (e.g., imitation–play, mastery motivation) than of behavioral and temperamental characteristics. For instance, Carter, Briggs-Gowan, Jones, and Little (2003) argued that social competence reflects the portrayal of age-appropriate social skills and is not simply a lack of problem behavior. Parent competence beliefs, then, may be aligned more closely with aspects of children's behavior that might be modified by early intervention services.

We also found that Parent Outcome Expectations, but not Parent Competence, scores were negatively related to reports of children's internalizing behaviors. Several researchers have studied children's internalizing problems as they relate to PSE in families of children without disabilities (e.g., Johnston & Mash, 1989; Mash & Johnston, 1983). In the development literature for children with and without disabilities, however, externalizing problems and dysregulation have received the majority of attention possibly because of the evident stress that these types of problems present for parents. Therefore, it remains unclear whether the relation found in the current research was spurious or whether there was a true relation between children's internalizing behaviors and mothers' outcome expectations. Several items in the Parent Outcome Expectations factor may have contributed to this particular finding (i.e., 12 and 20). Specifically, mothers may be more likely to view internalizing behaviors as genetically based (outcome expectations) and not necessarily a function of their personal parenting capabilities (parent competence).

With respect to the relation between EIPSES scales and developmental scale scores on the BDI (Newborg et al., 1988), there were no consistent findings. Receptive language was the only developmental domain for which there was any statistically significant relation to the EIPSES, such that higher total and Outcome Expectations subscale scores were related to higher scores on children's receptive language. To our knowledge, few studies have examined the link between child language and parental feelings of self-efficacy (cf. Harty, Alant, & Uys, 2007); hence, these findings are tentative and deserve further examination.

Although few patterns have emerged from previous studies across a range of populations and PSE measures, the present findings provide additional information to help tease out the complexities of the relation between EI PSE and a variety of demographic variables. In our study, we found that there were differences in levels of EI PSE beliefs depending on level of education and ethnicity. Our findings were in support of researchers who have suggested that lower levels of parental education are related to less favorable caregiving beliefs and dampened feelings of PSE (e.g., Hoover-Dempsey, Bassler, & Brissie, 1992; Huston, McLoyd, & Garcia Coll, 1994; Luster & Rhoades, 1989; Seefeldt, Denton, Galper, & Younoszai, 1999).

It is interesting to note that in our sample, individuals with associate's degrees had lower feelings of EI PSE than not only those with college and graduate school degrees but also those with only some post–high school education and no degree. It is possible that those who had some post–high school education had received more years of education than those with only an associate's degree, which typically can be obtained in 2 years. Determining the number of years of education or one's profession, rather than the degree level obtained, might shed more light on this finding.

Although we did find that Hispanic mothers had lower levels of parent outcome expectations efficacy than White mothers did, this difference is probably not straightforward and is likely complex in nature. In their sample of low-income, Head Start families, Machida and colleagues (2002) found that the relation between family stress and PSE was moderated by ethnicity, where family stress had a stronger affect on Mexican American than Anglo American mothers' parenting self-efficacy beliefs. Cultural differences may affect both a parent's feelings of self-efficacy and the way in which a parent interacts with his or her child with a disability. There is some evidence that the meaning of the disability may change depending on ethnic background (see Hanson, 1992, for a review). Furthermore, other investigators have determined that parent–child interaction style may differ depending on ethnicity (e.g., Garcia Coll, 1990; Laosa, 1980). Taken together, the current and past findings regarding self-efficacy and ethnicity reveal some important issues to consider and merit further attention.

We found no relations between EI PSE and child or parent age, number of children in the family, household income, or the parent-reported severity of the child's disability. In contrast, in their study comparing several domain-general self-efficacy measures to a domain-specific measure among parents of typically developing children, Coleman and Karraker (2000) obtained statistically significant positive relations between general measures of parenting self-efficacy and child age, mother's education, family income, and experience with children. These findings differed, however, depending on whether a domain-general or domain-specific measure was used. In support of our findings, the number of children in the family, mother's age, and marital status were not related to PSE for any of the scales the authors tested. Furthermore, in their study of children with clinically referred behavior disorders, Scheel and Rieckmann (1998) found that higher levels of PSE were related to being employed and being a single parent. PSE was not related, however, to parental education or number of children in the family. Because of discrepant findings across studies, the relationship of PSE and demographic variables remains unclear. Future replications of the associations of domain- and task-specific PSE to demographic factors may further our understanding of the relationship. In addition, it is important to note that the magnitude of these relations was low to moderate; therefore, these results should not be overstated.

Limitations and Future Directions

There are limitations to this study that should be noted. First, similar to other studies involving PSE beliefs, the present investigation was restricted to concurrent reports of parental feelings of competence and child social, emotional, and developmental functioning (cf. Coleman & Karraker, 1998). Thus, causal inferences of directional effects between EIPSES scores and children's social, emotional, and developmental scores should not be made. Second, although the relations between children's internalizing, externalizing, dysregulation, and competence behaviors with self-efficacy was anticipated, this relation may also be a function of shared bias in mothers' reports of both their own feelings of self-efficacy and their perceptions of children's social and emotional behaviors. In line with this idea, researchers have noted that parent reports of child behaviors tend to be dependent on past experiences and interactions, both positive and negative, such that experiences and interactions in the past bias a parent's current perceptions of his or her child's behaviors (e.g., Holleran, Littman, Freund, & Schmaling, 1982).

Factor coefficients and internal consistency reliability of scores for the Parent Outcome Expectations factor reached acceptable levels; however, the Parent Competence factor had only average score reliability. This low score reliability was likely due, in part, to the small number of items in that factor ($n = 4$). Because only 37% of the variance was explained by the two factors, further verification of a two-factor structure should be replicated with larger and more diverse samples to determine whether a similar factor structure is obtained. Furthermore, because the assumption of a normal distribution was not met, the factor solutions and descriptive analyses were likely degraded (Tabachnick & Fidell, 2007). Hence, the two-factor scale should be used with some caution at this initial phase of scale development before further psychometric assessment of the EIPSES has been conducted. Nevertheless, with an acceptable level of score reliability in the current sample, the EIPSES overall scale is a useful option for early interventionists.

Overall, parents had high levels of EI self-efficacy and seemed to feel confident and competent in their abilities to exert control over their children's early intervention outcomes and promote their children's development. The restricted range in scores, however, may have decreased the total variance in scores and, thus, the reliability coefficient (Anastasi & Urbina, 1997). Furthermore, the sample consisted of a relatively homogeneous group of White mothers from middle-income households, and sampling was restricted to a sample of convenience in three southwestern states. Therefore, the EIPSES may not necessarily generalize to other populations of parents, nor is our sample representative of the general populations of parents of children receiving early intervention services. Replication of the EIPSES with different sampling frames will inform our understanding of the stability of the obtained parameter estimates.

There are also limitations regarding the sample for the present investigation. The roles of child gender and ethnicity were not thoroughly explored in our analyses. Although there were no mean differences on any of the study variables between boys and girls, and small differences between White and Hispanic groups, it is possible these variables may be moderators. For instance, it may be that the relation between children's social emotional behaviors and EI self-efficacy beliefs are different for boys than for girls or for Hispanic versus White individuals. Child sex and ethnicity are generally shown to have moderating effects in studies of parenting beliefs and behaviors (e.g., Crockenberg, 1986; Hudson et al., 2001; Machida et al., 2002). Therefore, in the future, researchers should examine the moderating affects of gender and ethnicity on PSE and EI PSE.

Despite examinations of potential differences between mothers' and fathers' cognitive appraisals in raising a child with a disability (e.g., Hastings & Brown, 2002; Sharpley, Bitsika, & Efremidis, 1997), scholars have noted a general lack of attention to differences between maternal and paternal cognitions in the existing literature (e.g., Bugental & Johnston, 2000). Nevertheless, some evidence exists that maternal and paternal self-efficacy may be differentially related to child and family outcomes (cf. Jones & Prinz, 2005), and future investigations should analyze the psychometric properties of the EIPSES with fathers.

Implications

Considerable attention to parenting self-efficacy has revolved around the idea that self-efficacy beliefs should be a target of parenting interventions. Highlighting the role of self-efficacy beliefs as a possible mechanism through which both parent well-being and parenting behaviors become apparent, researchers have recently begun to examine the ways in which self-efficacy beliefs have a direct impact on parenting quality and skill as well as how those changes indirectly affect child behavior and development (Jones & Prinz, 2005). Researchers have suggested that efforts to increase caregiver self-efficacy through early intervention may positively affect parenting quality and, thus, child well-being (e.g., Izzo, Weiss, Shanahan, & Rodriguez-Brown, 2000). By better understanding how parental perceptions contribute to family well-being, interventionists might effectively target change in the caregiver's cognitive environment and potentially change parenting practices and responses to contextual and family stress.

Jones and Prinz (2005) reviewed a number of studies linking higher levels of PSE to improved child functioning, and their findings may lend support to the notion that PSE and child outcomes might be related by means of self-efficacy's potential influence on parenting quality. In addition to its role as an outcome of EI services, PSE has been shown to mediate the relations between child characteristics and both parenting quality and parental well-being. Specifically, child behavior problems have been related to lower levels of parenting quality and decreases in parent well-being through their negative relation with parenting self-efficacy beliefs (e.g., Cutrona & Troutman, 1986; Donovan et al., 1990; Teti & Gelfand, 1991). These findings support the idea that child and family characteristics may affect parenting quality via their more indirect path through parents' self-efficacy beliefs (Teti, O'Connell, & Reiner, 1996) and have implications for the utility of interventions that focus on enhancing parenting self-efficacy.

The potential implications of the present investigation, as well as previous research efforts by others, indicate that EI PSE might be an important outcome in and of itself as well as a precursor to later child and family outcomes. By altering parental perceptions of competence, actual parenting competence may be fostered and, thus, child outcomes promoted. Developing assessments to measure self-efficacy beliefs effectively with families of children with disabilities is an important step in furthering the study of early intervention. In addition, these instruments might assist researchers and interventionists as they attempt to understand how perceptions may influence parenting behaviors and child outcomes. Although there are many other efficacy-related measures in existence, the EIPSES might be valuable for use in identifying strengths and weaknesses regarding mothers' perceived abilities in promoting the development of their children who are receiving early intervention services, information that might then be used as targeted changes in the families' intervention programs.

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